

Final Project Options

Due: May 6, 2026; 10:30am; 779 Davis

This project will comprise 40% the final exam grade.

Your project report should be typed. There are two options for the final project one is a FEA programming project and the other is an analysis project.

For both projects your report should contain the following sections: (a) an executive summary, (b) introduction, (c) modeling approach, (d) analysis of results and accuracy assessment, (f) appendix with input files for final analysis, (g) appendix with auxiliary analyses performed for simple test cases, (h) appendix with other back-up material. Appendices should be referred to within the body of your report. All plots and tables should be accompanied by explanatory text, proper captions, legends, etc. **Simply put, be professional.**

I estimate the amount work required for both projects to be roughly equivalent. So the project that you choose is really a matter of personal taste. You should get started on your project, no later than the first day after spring break.

The final project is intended to be a measure of the abilities that **you** have gained over the term; thus, these are not group projects, but you are of course free to discuss with each other.

Note that every **Friday in April**, we will go around the class and everyone will be expected to make a brief 1 or 2 minute update on what they have done so far in the project. This will also be an opportunity to talk as a group to discuss any issues that you may encounter.

All reports should be accompanied by the following statement and a signature attesting to the statement:

"Honor Code: This report represents my own work and not that of anyone else. The writing is my own, as are all input files, programs, analysis results, and interpretations. I have taken an active part in seeing to it that others as well as myself uphold the spirit and letter of this Honor Code."

(Signature)

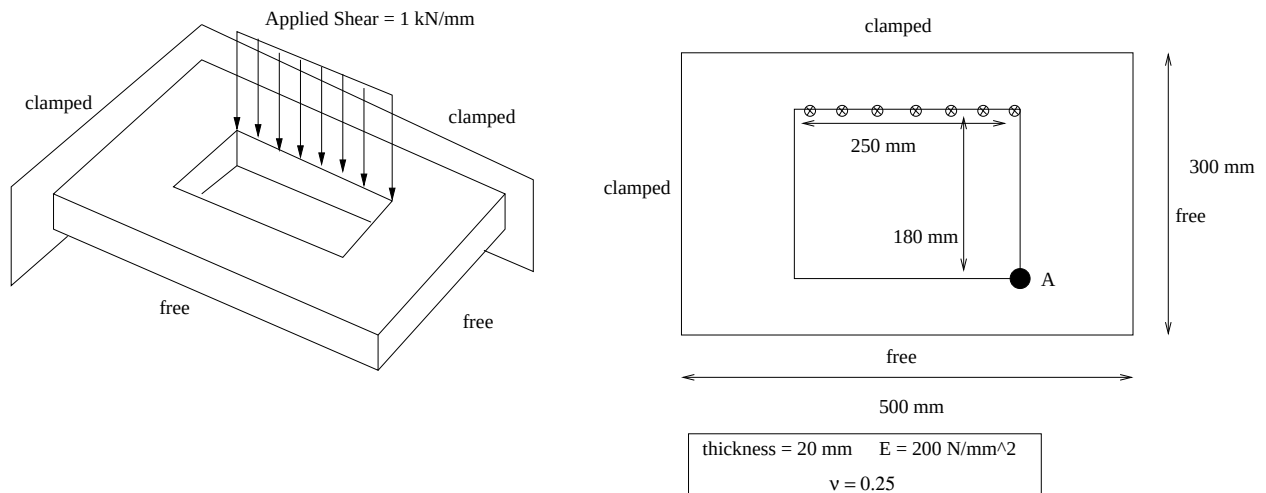
Project option 1 is a programming project.

You are to write a finite element program that solves a particular problem using a single element type – the heterosis plate element for a shear deformable isotropic elastic plate. This element will be covered in class the first week after spring break; it is a Q8 element for u_3 and a Q9 for θ_1 and θ_2 with a 3×3 quadrature rule on the bending part of the stiffness terms and a 2×2 quadrature on the shear part of the stiffness terms. Note that you can get started without the lectures by programming infrastructure for generating your mesh, routines for evaluating shape functions and their derivatives, etc.

One source among many for the plate theory is T.J.R. Hughes *The Finite Element Method*, Prentice-Hall (1987), Chapter 5 up through §5.3.7 where he explicitly talks about the Heterosis element. You can also read about thick-plates (shear deformable plates) in ZT&G Chapter 19.

Note that there are some clever tricks for implementing this element but I think it is better to perform a straightforward implementation when doing things for the first time.

You are then to use this plate element to determine the transverse deflection at point A of the plate structure shown in the figure below. (The cut-out is centered.) Your report should include a summary of the plate equations and the FEA equations, it should include test cases from your development phase, and definitely convergence studies. You are welcome to also use FEAPpv to determine baseline solutions to compare against.



Project option 2 is an analysis project.

You are to study the Sleipner deep water structure failure that occurred in 1991. This accident is likely the largest monetary loss ever due to a finite element analysis mistake. In today's dollars, it was an approximately \$1.8B loss. Thankfully no one lost their life in the accident.

I have assembled a number of reports and articles that were written about the disaster.¹ You are welcome to search out others that may help you. Your task is to re-analyze the structural system and provide an assessment of why the structure failed and sunk – confirming, rebutting, and/or extending the statements made in the reports and articles you have been given or have located yourself. That analysis necessarily requires that you perform an FEA study (using FEAPpv) of the critical component and/or components. This analysis needs to be carefully executed and properly double checked, just as you would do in a professional engineering office. Your report should present the models that you use and the modeling assumptions that you are making; make sure to include simple “hand computations” that you have used to ensure your results make sense, all test cases, convergence studies, etc.

Note, this problem is typical of many real life questions. You are faced with a problem that is only partially defined and you will have to make and justify some modeling assumptions; some have been conveniently suggested in the reports and articles, while others are left unstated. Further you are constrained by cost, time, and resource issues to treat the problem.

¹The reports and articles include two articles in German, 6-7. I include these, as they have some nice images. The text is not so important as it is all covered in 1-5; though you can always put them through Google translate if you are curious.